

Experiment No. 2

Aim: Design a Login Screen Using UI Components and Event Handlers

Apparatus / Software Required :

Hardware	- Laptop(ASUS Vivobook 16x) - Minimum 8 GB RAM - AMD Ryzen 5 5600H with Radeon Graphics Processor - Internet Connection
Software	- Windows 11 Operating System - Android Studio (Quail) - Java Development Kit (JDK) - Android SDK - Android Emulator (Pixel 7) - Gradle Build System

Theory

A login screen is used to authenticate a user before allowing access to an application. In Android, different UI components such as TextView, EditText, Button and ImageView are used to create the login interface.

The EditText component is used to take username/email and password from the user. A Button is used to perform an action when the user clicks it. Event handlers are used to handle these button click events.

In this experiment, a simple and attractive login screen was designed using XML. Java event handling was used to check the entered username and password and display an appropriate message such as Login Successful or Invalid Username or Password.

Implementation Steps:

Step 1: Create Android Project

1. Open Android Studio.
2. Click on New Project.
3. Select Empty Views Activity.
4. Select Java as the programming language.
5. Select the required minimum SDK.
6. Enter the project name and click Finish.

Step 2: Design Login Screen

1. Open activity_main.xml.
2. Add a suitable parent layout such as ConstraintLayout.
3. Add an ImageView for the application logo.
4. Add an EditText for username/email.
5. Add another EditText for password.
6. Add a Login Button.
7. Add Forgot Password TextView.
8. Add an OR separator.
9. Add a button for Login with Facebook.
10. Add a Sign Up TextView.

Step 3: Add Event Handling

1. Open MainActivity.java.
2. Connect the XML components using findViewById().
3. Add a click listener to the Login button.
4. Read the username/email and password entered by the user.
5. Check whether the entered values are valid.
6. Display Login Successful for correct credentials.
7. Display Invalid Username or Password for incorrect credentials.

Step 4: Run the Application

1. Start the Android Emulator.
2. Select the created virtual device.
3. Click the Run button.
4. Enter test username and password.
5. Click the Login button.
6. Check the displayed result.

Files Used and Code Changes

1. activity_main.xml

The login screen user interface was designed using the XML layout file activity_main.xml.

Changes Made:

- Added application logo using ImageView.

- Added username/email input field using EditText.
- Added password input field using EditText.
- Added a second TextView displaying:
 - “Developed by RAJ SIRSODE”
- Added **Login** button.
- Added **Forgot Password** option.
- Added **OR** separator.
- Added **Login with Facebook** button.
- Added **Sign Up** option.
- Applied background, text size, font style and margins.
- Used ConstraintLayout constraints to properly position the components.

2. MainActivity.java

Java code was used to handle the events of the login screen.

Changes Made:

- Initialized UI components using findViewById().
- Added click event for the Login button.
- Read username/email and password using getText().
- Added validation for login credentials.
- Used Toast to display the result.
- Displayed **Login Successful** when the credentials were correct.
- Displayed **Invalid Username or Password** when the credentials were incorrect.

Components Used

Component	Purpose
ConstraintLayout	Used as the main layout to arrange UI components
ImageView	Used to display the application/logo image
TextView	Used for labels, Forgot Password, OR and Sign Up
EditText	Used to enter username/email and password
Button	Used to perform the Login action
Toast	Used to display login result messages

Important XML Attributes Used

XML Attribute	Purpose
android:layout_width	Defines the width of a component
android:layout_height	Defines the height of a component
android:text	Sets the text displayed by a component
android:textSize	Sets the size of the text
android:textColor	Sets the color of the text
android:textStyle	Makes text bold or changes its style
android:background	Sets the background of a component
android:hint	Displays a hint inside an input field
android:inputType	Defines the type of input, such as text or password
android:padding	Adds space inside a component
android:layout_margin	Adds space around a component
app:layout_constraintTop_toTopOf	Positions a component relative to the top
app:layout_constraintBottom_toBottomOf	Positions a component relative to the bottom
app:layout_constraintStart_toStartOf	Aligns a component to the start
app:layout_constraintEnd_toEndOf	Aligns a component to the end

Important Java Methods Used

1. onCreate()

- The onCreate() method is called when the activity is created. It is used to initialize the activity and load the XML layout.

@Override

protected void onCreate(Bundle savedInstanceState)

2. setContentView()

- This method connects the XML layout file with the Java Activity.

setContentView(R.layout.activity_main);

3. findViewById()

- This method is used to connect the XML UI components with Java variables.

```
EditText username = findViewById(R.id.etUsername);
```

4. setOnClickListener()

- This method is used to perform an action when the user clicks a button.

```
btnLogin.setOnClickListener(v -> {  
    // Login operation  
});
```

5. getText()

- This method is used to get the value entered by the user in an EditText.

```
String username = etUsername.getText().toString();
```

6. Toast.makeText()

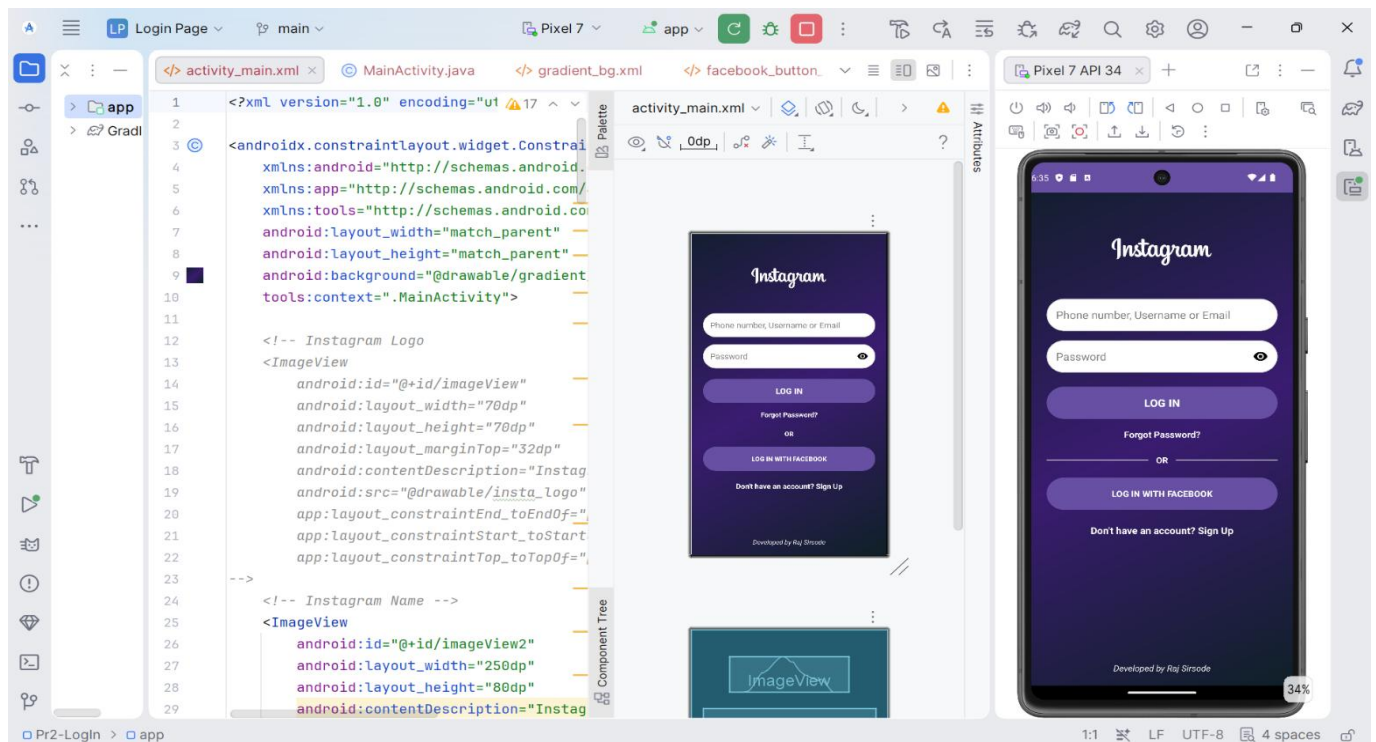
- This method is used to display a short message to the user.

```
Toast.makeText(this, "Login Successful", Toast.LENGTH_SHORT).show();
```

Errors Faced and Troubleshooting

Error Faced	Solution
Gradle Sync Failed	Checked Gradle and dependency versions and performed Gradle Sync again
Incompatible Gradle JVM	Selected a compatible JVM 17
Emulator setup problem	Created a Pixel virtual device with API 34
UI components were not properly aligned	Added proper ConstraintLayout constraints
Login button was not responding	Checked setOnClickListener() and button ID
Incorrect login message	Checked username and password validation conditions
Password was visible while entering	Used appropriate inputType for password
Layout preview was not displayed correctly	Fixed XML errors and performed project sync

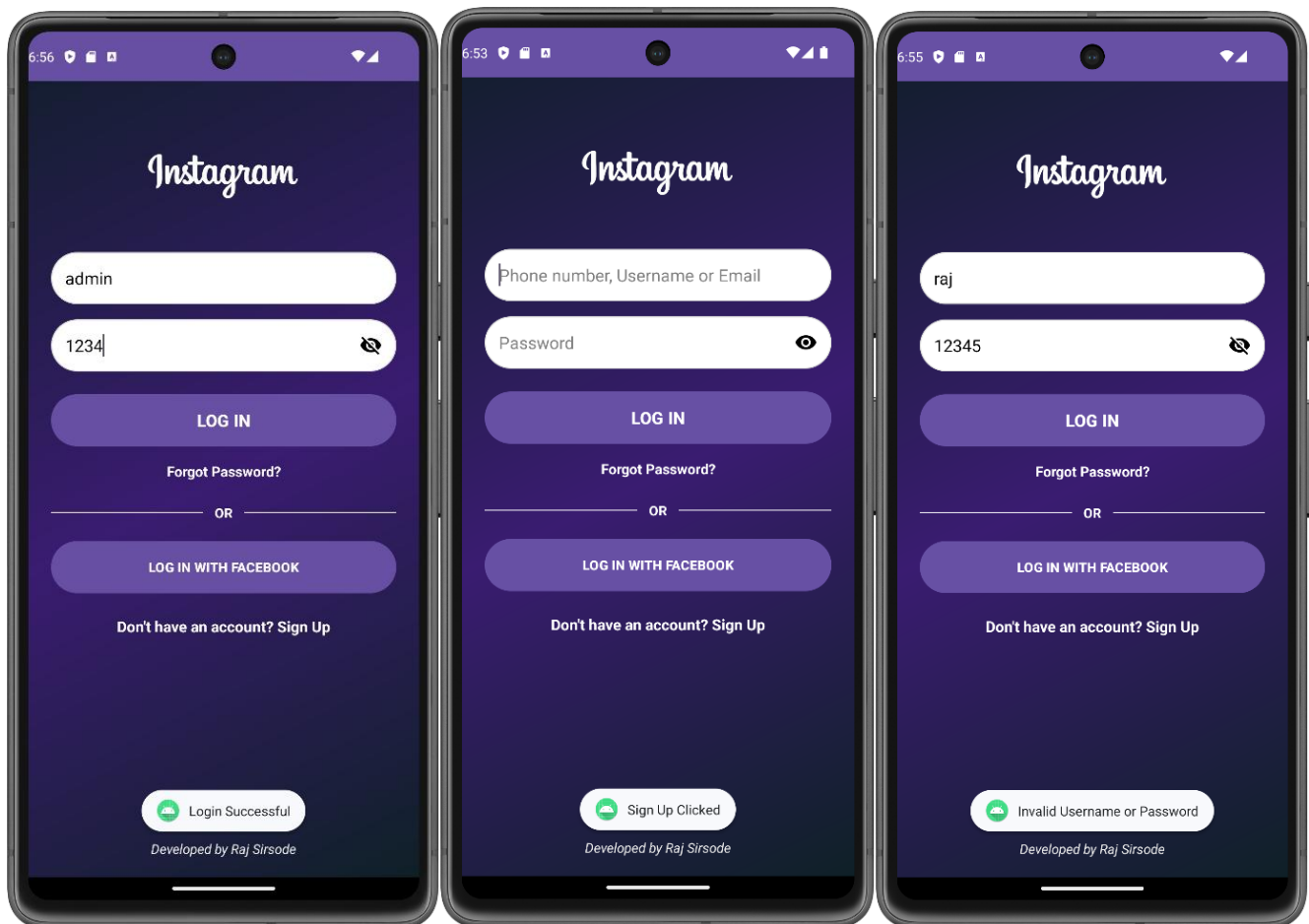
Output



Login

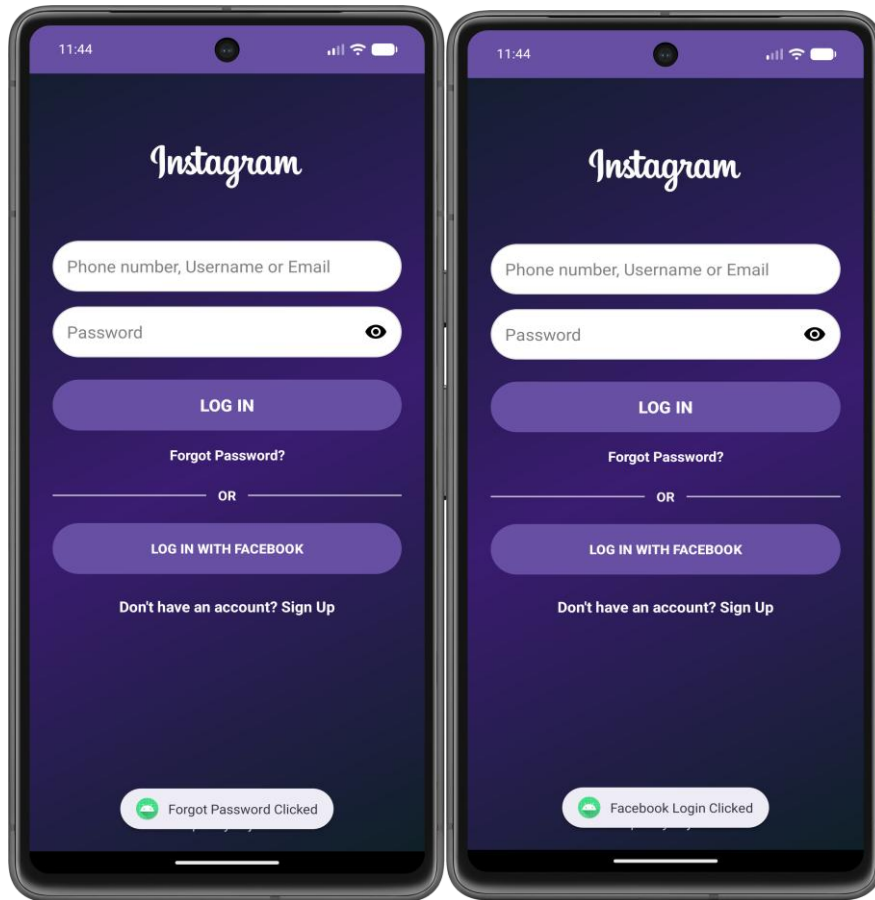
Sign Up

Invalid user & pass



Forgot Pass

Facebook Login



Conclusion

The login screen was successfully designed using Android UI components and XML. Different components such as **EditText**, **Button**, **TextView** and **ImageView** were used to create the interface. Java event handlers were implemented to handle the Login button click and validate the entered credentials. The application successfully displayed appropriate messages such as **Login Successful** and **Invalid Username or Password**. Through this experiment, I learned how to design an Android user interface and handle user events using Java.

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